

Determining the Equation of a Regression Line

How Has Brain Volume Changed Over Time in the Hominin Lineage?

The hominin taxon includes *Homo sapiens* and about 20 extinct species that are thought to represent early relatives of humans. Researchers have found that the brain volume of the earliest hominins ranged between 300 and 450 cm³, similar to the brain volume of chimpanzees. The brain volumes of modern humans range between 1,200 and 1,800 cm³. In this exercise, you'll examine how mean brain volume changed over time and across various hominin species.



How the Study Was Done In this table, x_i is the mean age of each hominin species, and y_i is the mean brain volume (cm³). Ages with negative values represent millions of years before the present (which has an age of 0.0).

Hominin Species	Mean age (millions of years; x_i)	$x_i - \bar{x}$	Mean Brain Volume (cm ³ ; y_i)	$y_i - \bar{y}$	$(x_i - \bar{x}) \times (y_i - \bar{y})$
<i>Ardipithecus ramidus</i>	-4.4		325		
<i>Australopithecus afarensis</i>	-3.4		375		
<i>Homo habilis</i>	-1.9		550		
<i>Homo ergaster</i>	-1.6		850		
<i>Homo erectus</i>	-1.2		1,000		
<i>Homo heidelbergensis</i>	-0.5		1,200		
<i>Homo neanderthalensis</i>	-0.1		1,400		
<i>Homo sapiens</i>	0.0		1,350		

Data from Dean Falk, Florida State University, 2013.

INTERPRET THE DATA

How did the brain volume of hominin species change over time? In particular, is there a linear (straight-line) relationship between brain volume and time?

To find out, we'll perform a linear regression, a technique for determining the equation for the straight line that provides a "best fit" to a set of data. Recall that the equation for a straight line between two variables, x and y , is

$$y = mx + b$$

In this equation, m represents the slope of the line, while b represents the y -intercept (the point at which the straight line crosses the y -axis). When $m < 0$, the line has a negative slope, indicating that the values of y become smaller as values of x become larger. When $m > 0$, the line has a positive slope, meaning that the

values of y become larger as values of x become larger. When $m = 0$, y has a constant value (b).

The correlation coefficient, r , can be used to calculate the values of m and b in a linear regression:

$$m = r \frac{s_y}{s_x} \text{ and } b = \bar{y} - m\bar{x}.$$

In these equations, s_x and s_y are the standard deviations of variables x and y , respectively, while $\bar{x}$ and $\bar{y}$ are the means of those two variables. (See the Scientific Skills Exercise for Chapter 32 for more information about the correlation coefficient, mean, and standard deviation.)

- Calculate the means ($\bar{x}$ and $\bar{y}$) from the $n = 8$ data points in the table. Next, fill in the $(x_i - \bar{x})$ and $(y_i - \bar{y})$ columns in the data table, and use those results to calculate the standard deviations s_x and s_y .
- As described in the Scientific Skills Exercise for Chapter 32, the formula for a correlation coefficient is

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)(s_x s_y)}$$

Fill in the column in the data table for the product $(x_i - \bar{x}) \times (y_i - \bar{y})$. Use these values and the standard deviations calculated in question 1 to calculate the correlation coefficient r between the brain volume of hominin species (y) and the ages of those species (x).

- Based on the value of r that you calculated in question 2, describe in words the correlation between mean brain volume of hominin species and the mean age of the species.
- (a) Use your calculated value of r to calculate the slope (m) and the y -intercept (b) of a regression line for this data set. (b) Graph the regression line for the mean brain volume of hominin species versus the mean age of the species. Be careful to select and label your axes correctly. (c) Plot the data from the table on the same graph that shows the regression line. Does the regression line appear to provide a reasonable fit to the data?
- The equation for a regression line can be used to calculate the value of y expected for any particular value of x . For example, suppose that a linear regression indicated that $m = 2$ and $b = 4$. In this case, when $x = 5$, we expect that $y = 2x + 4 = (2 \times 5) + 4 = 14$. Based on the values of m and b that you determined in question 4, use this approach to determine the expected mean brain volume for a hominin that lived 4 million years ago (that is, $x = -4$).

- The slope of a line can be defined as $m = \frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are the coordinates of two points on the line. As such, the slope represents the ratio of the rise of a line (how much the line rises vertically) to the run of the line (how much the line changes horizontally). Use the definition of the slope to estimate how long it took for mean brain volume to increase by 100 cm³ over the course of hominin evolution.

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

the neck. Compared to the australopiths, *H. habilis* had a shorter jaw and a larger brain volume, about 550–750 cm³. Sharp stone tools have also been found with some fossils of *H. habilis* (the name means "handy man").

Fossils from 1.9 to 1.5 million years ago mark a new stage in hominin evolution. A number of paleoanthropologists recognize these fossils as those of a distinct species, *Homo ergaster*. *Homo ergaster* had a substantially larger brain than